

Digital SAT Math — Formula Cheat Sheet

All 4 domains · 44 questions · two modules · 70 minutes · certpathpublishing.store

ALGEBRA

Lines & Linear Equations

$$y = mx + b$$

- **Slope:** $m = (y_2 - y_1)/(x_2 - x_1)$
- **Point-slope:** $y - y_1 = m(x - x_1)$
- **Standard:** $Ax + By = C \rightarrow \text{slope} = -A/B$
- **Parallel:** equal slopes · **Perp:** $m_1 \cdot m_2 = -1$

Systems & Inequalities

- Solve 2x2 by **substitution** or **elimination**.
- **No solution:** same slope, different intercept (parallel).
- **Infinite solutions:** identical lines.
- Flip the inequality sign when $\times/\div$ by a negative.

ADVANCED MATH

Quadratics

$$x = \left[\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \right]$$

- **Vertex form:** $y = a(x - h)^2 + k \rightarrow \text{vertex } (h, k)$
- **Axis / vertex x:** $x = -b/2a$
- **Discriminant $b^2 - 4ac$:** >0 two real · $=0$ one · <0 none
- **Sum of roots** $= -b/a$ · **Product** $= c/a$

Exponents & Functions

- $x^a \cdot x^b = x^{a+b}$ · $x^a/x^b = x^{a-b}$ · $(x^a)^b = x^{ab}$
- $x^0 = 1$ · $x^{-n} = 1/x^n$ · $x^{1/n}$ = the n-th root of x
- **Growth / decay:** $y = a(1 \pm r)^t$ (r = rate per period)
- **Composition** $f(g(x))$: evaluate the inner function first.

PROBLEM-SOLVING & DATA

Percent, Ratio, Rate

- **Percent change:** $(\text{new} - \text{old})/\text{old} \times 100$
- **Percent of:** part = percent (as decimal) $\times$ whole
- **After x% off:** price = original $\times (1 - x)$
- **Proportion:** $a/b = c/d \rightarrow \text{cross-multiply}$

Statistics & Probability

- **Mean** = sum $\div$ count · **Median** = middle value
- Outliers pull the **mean**; the **median** resists them.
- **P(event)** = favorable $\div$ total outcomes
- Bigger random sample $\rightarrow$ smaller margin of error ($\propto 1/\sqrt{n}$).

GEOMETRY & TRIGONOMETRY

Triangles & Circles

- **Pythagorean:** $a^2 + b^2 = c^2$
- **45-45-90:** $x : x : x\sqrt{2}$ · **30-60-90:** $x : x\sqrt{3} : 2x$
- **Circle:** $A = \pi r^2$ · $C = 2\pi r$
- **Circle equation:** $(x - h)^2 + (y - k)^2 = r^2$
- Similar triangles: areas scale by (ratio) 2 .

Trig — SOHCAHTOA

$$\sin = O/H \quad \cos = A/H \quad \tan = O/A$$

- **Cofunction:** $\sin \theta = \cos(90^\circ - \theta)$
- Common triples: 3-4-5, 5-12-13, 8-15-17.
- Angles in a triangle sum to 180° .

🖨️ **Desmos & Bluebook:** The built-in Desmos calculator is on every math question — **graph it** to find intercepts, intersections, and vertices instead of solving by hand. **Type-in (student-produced) answers:** enter a fraction (5/2) or decimal (2.5) — **never a mixed number**; negatives and decimals are accepted; no %, \$, or commas. **On the provided reference sheet:** areas, volumes, and the special-right-triangle ratios. **Memorize** what is NOT given: the quadratic formula, slope, and exponent rules.